DIGITAL FORENSICS LAB REPORT

Topic: Five-Stage Digital Forensic Investigation Lifecycle Program

1. Aim

To demonstrate the five-stage digital forensic investigation lifecycle (Identification, Preservation, Collection, Analysis, and Reporting) by writing a structured Python program that models and logs forensic evidence and procedural workflows.

2. Algorithm

1. **Start** the program.
2. Define a structured data collection representing the five core phases of digital forensics alongside their objectives, activities, and generated artifacts.
3. Initialize investigation metadata including Case ID, Investigator Name, and Timestamp.
4. Format a structured forensic report template.
5. Iterate through each stage sequentially and log:
    - Stage Name
    - Primary Objective
    - Key Investigative Activities
    - Artifacts / Trace Evidence Gathered
6. Display or export the structured report output.
7. **Stop** the program.

3. Code

```
import time

def generate_forensic_report():
    # Forensic Investigation Metadata
    case_number = "DF-2026-0091"
    investigator = "Forensic Analyst"
    timestamp = time.ctime()

    # The Five-Stage Digital Forensic Lifecycle Data
    lifecycle_stages = [
        {
            "stage": "1. Identification",
            "objective": "Detect incident, identify potential evidence, and scope the investigation.",
            "activities": "Incident detection, pinpointing affected systems and devices.",
            "evidence": "System logs, network traffic captures, physical media"
        },
```

```
        {
            "stage": "2. Preservation",
            "objective": "Secure the crime scene and isolate digital assets to prevent
contamination.",
            "activities": "Write-blocking, network isolation, chain-of-custody
documentation.",
            "evidence": "Chain of custody log, physical tamper-evident bags"
        },
        {
            "stage": "3. Collection",
            "objective": "Acquire bit-stream copies of evidence while maintaining
integrity.",
            "activities": "Forensic imaging, RAM capture, SHA-256 hash generation.",
            "evidence": "Forensic image files (.E01 / .raw), memory dumps"
        },
        {
            "stage": "4. Analysis",
            "objective": "Examine extracted data to reconstruct events and identify
root causes.",
            "activities": "Timeline reconstruction, artifact extraction, file
carving.",
            "evidence": "Registry hives, browser history, event logs, deleted files"
        },
        {
            "stage": "5. Reporting",
            "objective": "Present clear, legally admissible findings to stakeholders
or court.",
            "activities": "Documentation of methodology, forensic summary, expert
report.",
            "evidence": "Final forensic report, verified hash logs"
        }
    ]

    # Print Report Header
    print("=" * 65)
    print("        DIGITAL FORENSIC INVESTIGATION LIFECYCLE REPORT        ")
    print("=" * 65)
    print(f"Case ID      : {case_number}")
    print(f"Investigator : {investigator}")
    print(f"Generated On : {timestamp}")
    print("-" * 65)

    # Output details for each stage
    for item in lifecycle_stages:
        print(f"\n[ STAGE ] {item['stage']}")
        print(f" Objective  : {item['objective']}")
        print(f" Activities : {item['activities']}")
        print(f" Artifacts  : {item['evidence']}")
        print("-" * 65)
```

```
if __name__ == "__main__":
    generate_forensic_report()
```

4. Output

```
=================================================================
        DIGITAL FORENSIC INVESTIGATION LIFECYCLE REPORT
=================================================================
Case ID      : DF-2026-0091
Investigator : Forensic Analyst
Generated On : Tue Jul 28 08:24:00 2026
-----------------------------------------------------------------

[ STAGE ] 1. Identification
 Objective   : Detect incident, identify potential evidence, and scope the investigation.
 Activities : Incident detection, pinpointing affected systems and devices.
 Artifacts   : System logs, network traffic captures, physical media
-----------------------------------------------------------------

[ STAGE ] 2. Preservation
 Objective   : Secure the crime scene and isolate digital assets to prevent contamination.
 Activities : Write-blocking, network isolation, chain-of-custody documentation.
 Artifacts   : Chain of custody log, physical tamper-evident bags
-----------------------------------------------------------------

[ STAGE ] 3. Collection
 Objective   : Acquire bit-stream copies of evidence while maintaining integrity.
 Activities : Forensic imaging, RAM capture, SHA-256 hash generation.
 Artifacts   : Forensic image files (.E01 / .raw), memory dumps
-----------------------------------------------------------------

[ STAGE ] 4. Analysis
 Objective   : Examine extracted data to reconstruct events and identify root causes.
 Activities : Timeline reconstruction, artifact extraction, file carving.
 Artifacts   : Registry hives, browser history, event logs, deleted files
-----------------------------------------------------------------

[ STAGE ] 5. Reporting
 Objective   : Present clear, legally admissible findings to stakeholders or court.
 Activities : Documentation of methodology, forensic summary, expert report.
 Artifacts   : Final forensic report, verified hash logs
-----------------------------------------------------------------
```